

Nonnegative and positive factorizations

Additivity for Cartesian-product slack matrices

Difficulty: extreme **Importance:** interesting to the community

Status: Partially resolved

Rating rationale: Extreme because additivity across arbitrary polytopes is a general structural barrier for extension complexity; community importance links optimal factorizations with compositional linear optimization models.

Area: nonnegative rank and polyhedral optimization

Reviewed continuation — 13 September 2026

Sidney Holden (Center for Computational Biology, Flatiron Institute, Simons Foundation) supplies a computer-assisted proof that $\text{xc}(P \times Q) = 12$ whenever both factors are polygons of extension complexity six. The [fourth-round report](#), main theorem and geometry-to-Boolean reduction, reduces this statement to ten exact LRAT refutations. The broader inherited all-polytopes corollary in the package was not independently re-audited in this submission.

Status remains **Partially resolved**: arbitrary higher-complexity factors and unrestricted Cartesian-product additivity are not settled.

The [independent informal Codex AI-agent review](#) records its accepted scope and checks. [Submission, original package, reproduction and verified affiliation](#). This is not external human peer review or formal verification; no Lean verification was performed.

Context and notation

All factorizations are over the real numbers. For $X \in \mathbb{R}_{\geq 0}^{m \times n}$, define

$$\text{rank}_+(X) = \min\{r \geq 0 : X = WH, \quad W \in \mathbb{R}_{\geq 0}^{m \times r}, H \in \mathbb{R}_{\geq 0}^{r \times n}\}.$$

Problem statement

Let $P \subset \mathbb{R}^d$ and $Q \subset \mathbb{R}^e$ be full-dimensional polytopes with $d, e \geq 1$. Let $S \in \mathbb{R}_{\geq 0}^{m \times n}$ and $T \in \mathbb{R}_{\geq 0}^{p \times q}$ be their slack matrices: rows correspond to all facets in irredundant inequality descriptions, columns to all vertices, and each entry is the right-hand side minus the left-hand side of that facet inequality at that vertex. Construct $C \in \mathbb{R}_{\geq 0}^{(m+p) \times nq}$ by giving it one column for each pair (i, j) of vertices:

$$C[:, (i, j)] = \begin{pmatrix} S[:, i] \\ T[:, j] \end{pmatrix}.$$

Question

Is

$$\text{rank}_+(C) = \text{rank}_+(S) + \text{rank}_+(T)$$

always true? The hypotheses that S, T are polytope slack matrices are part of the question. Through the slack-factorization theorem, this also asks whether the minimum number of inequalities in an extended formulation is additive under Cartesian products.

References

Gillis, *Nonnegative Matrix Factorization*, §3.6.5, pp. 93–94. Hans Raj Tiwary, Stefan Weltge, and Rico Zenklusen, *Extension complexities of Cartesian products involving a pyramid*, Information Processing Letters **128** (2017), 11–13, introduction and main theorem.

Status check — 2026-09-10

Rechecked [Tiwary–Weltge–Zenklusen](#) and [Weltge’s 2022 workshop question](#), then searched for later product-additivity proofs and counterexamples. Equality is proved whenever one factor is a pyramid, a substantive portion of the displayed target. The unrestricted equality remains open in the latest explicit source located; no subsequent full resolution was found. The 2022 date limits the strength of the current-status evidence.